

ECE 537:
Advanced Data Analytics
Exercise #5:
Unbalanced Data Sets

Real-world data sets generally do not be well-balanced in terms of the numbers of available samples from each data class or behavioral set. For example, within cyber-security domains normal or non-adversarial events typically occur many orders of magnitude more frequently than attack events. In fault detection and medical domains it is not unusual to need to develop methods to detect events that may occur very rarely. Unbalanced data issues require that sufficient care must be taken in constructing data analysis systems and approaches that do not exhibit inherent biases towards classes with larger numbers of available samples.

Tasks:

- Use Matlab's Live Script feature to create a live script that does (and documents) the following work:
1. Use the methods you created in Exercise 1 to create a 3-dimensional data set consisting of 5 Gaussian classes, with arbitrary means and covariances, with the following number of samples drawn from each class:

Class	No. of Samples
Class A	750,000
Class B	190,000
Class C	45,000
Class D	12,000
Class E	3,000

Your data should be constructed such that all of the classes retain significant overlaps within one or two standard deviations of their means.

2. Construct randomly (uniformly) selected training and test data set from the composite of your constructed 5 class data sets, where the training data should constitute 10% of the overall data and the test data the remaining 90% of the data. Record the number of train data samples for each of the training set classes as well as the exact indexes of the training and test data samples.

3. Use the training data sets to build and a Bayes and neural network classifier for this data.
4. Apply the testing data to assess the performance of your constructed classifiers against each of the 5 classes. Explain why the classification performance is (or is not) consistent across each of Class A through E.
5. Discuss the similarities and differences that the unbalanced data sets have on the in sensitivity and robustness of your constructed classifiers and their training versus test set performances. Which types of classification approaches are more robust to unbalanced data sets (and why?) neural network type approaches, *i.e.*, non-parametric approaches or Bayes-like approaches, *i.e.*, parametric approaches?
6. Discuss the impacts that selecting random draws from unbalanced data sets can have on the composition of the resulting training data set(s). For each class, what will be the expected number of samples from the class drawn into the training data set and compute the probability of this occurring? How many possible draws from class E into the training set can occur with a probability of 99%? Compute how the range of possible draws changes as the size of class E is increased or decreased?
Hints: Which probability distribution could you use to model the likelihood(s) of drawing samples into the training set? How many standard deviations of this distribution are required to cover 99% of outcomes? , *i.e.*, fall into a 99% confidence interval.
7. Compare the performance of your classifiers to a second set of classifiers constructed from balanced training data, *i.e.*, generate balanced per-class data sets and retrain your classifiers.
8. Unbalanced data sets occur commonly in real-world data analysis problems *e.g.*, in cyber-security normal network traffic events typically outnumber malicious events by many orders of magnitude.
Discuss some of the possible approaches that could be applied to address the problem of unbalanced data. In general, there are three core approaches to addressing unbalanced data. Your discussion should include a detailed explanation of the pros and cons of each potential solution approach.

Note, in general, it may not be possible or feasible to simply collect sufficient data from each representative data class such that the training data with high likelihood is sufficiently balanced. Many real-world problems exist where data is innately unbalanced due to the nature of the problems space, *e.g.*, medical diagnostics, faculty detection, cyber-security, *etc.* .

9. Can unbalanced data sets be easily identified in real-world data analytics problems (why or why not)?

For unbalanced data, is it useful to be able to quantify the degree to which it is unbalanced. Is this often easy to do (why or why not)?

How would you classify the degree to which unlabelled data was unbalanced?

Could the balanced or unbalanced nature of a given data set change as you see (or collect) more data (why or why not)?